



ANALYTICA 2026

Scale with Intelligence

SAS CU Hackathon – Round 2

Team Name: *SHIKSHARTHI*

Team No: 50

Project Title:

From Entry-Level to Leadership: A Data-Driven Blueprint of Skills, Salaries, and Personality Traits for Success in Data Science Careers.

Submission Date:

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Introduction

Data Science has become one of the fastest-growing career fields in recent years. Companies depend on data-driven decisions, creating high demand for skilled professionals. Success in this field depends not only on technical skills but also on experience, communication abilities, and personality traits.

This study aims to analyze the journey of a Data Science professional from entry-level roles to leadership positions by identifying the key skills, salary factors, and personality traits that contribute to long-term career success.

Problem Statement

We were provided with multiple raw datasets related to Data Science jobs, analytics roles, salary trends, junior employee skill traits, and senior leadership personality traits. After reviewing and understanding the available data, we identified an opportunity to study the complete career journey of a Data Science professional.

The main challenge was to determine what factors truly contribute to long-term success in Data Science careers. Therefore, this project focuses on analyzing how technical skills, salary growth, experience, and personality traits influence career progression from entry-level roles to leadership positions.

Using a data-driven approach, this study presents “**From Entry-Level to Leadership**” to help students, professionals, recruiters, and organizations understand the key factors behind success in Data Science careers.

Approach / Objectives of the Problem

To address the selected problem statement, a structured data-driven approach has been adopted using the provided raw datasets related to Data Science jobs, salaries, skills, and personality traits. The study focuses on understanding the complete career journey of a Data Science professional from entry-level roles to leadership positions.

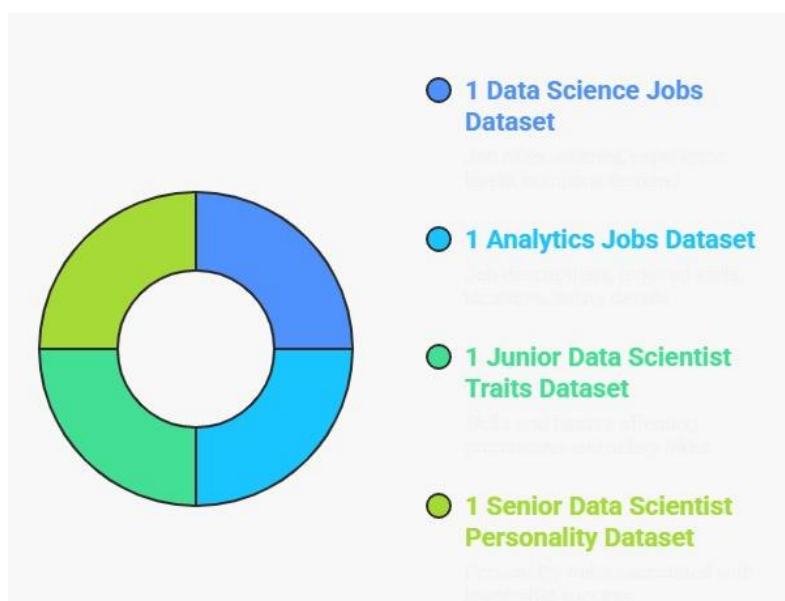


The key objectives of this project are: data derivation, data modification, Data Deduction, reductions

1. To identify the **most in-demand** technical and professional skills in Data Science careers.
2. To analyze **salary trends** based on job roles, experience, and required skills.
3. To determine the factors influencing **promotions and career growth** at junior levels.
4. To identify **personality traits** associated with success in leadership roles.
5. To develop a blueprint for career success titled “**From Entry-Level to Leadership.**”

Dataset Description

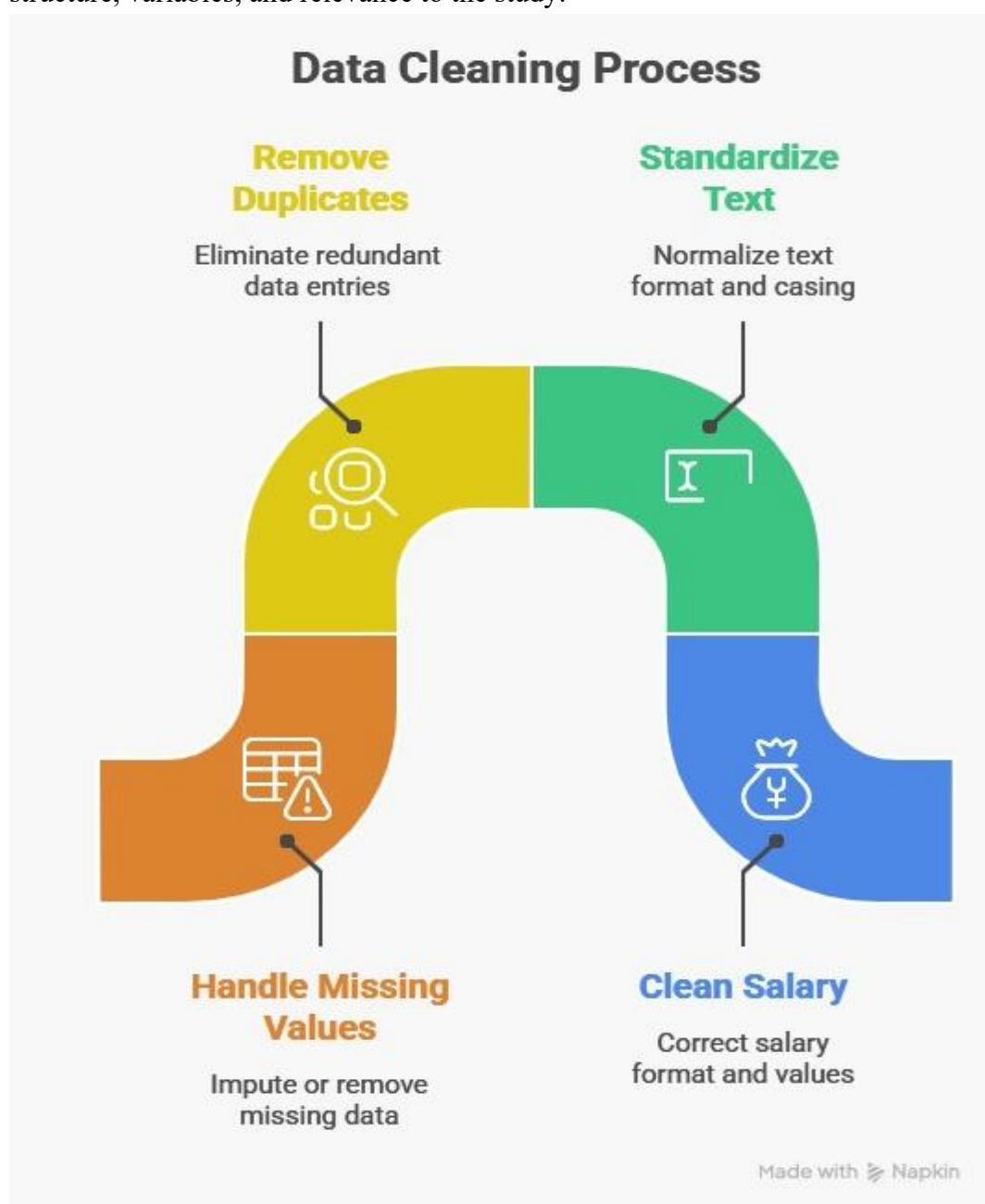
This project uses multiple raw datasets related to Data Science careers, job opportunities, salaries, skills, and personality traits. Each dataset provides insights into different stages of career growth from entry-level positions to leadership roles.



1. **Data Science Jobs Dataset** – Contains information on job roles, salaries, experience levels, and company demand.
2. **Analytics Jobs Dataset** – Includes job descriptions, required skills, locations, and salary details.
3. **Junior Data Scientist Traits Dataset** – Focuses on skills and factors affecting promotions and salary hikes.
4. **Senior Data Scientist Personality Dataset** – Contains personality traits associated with leadership success.

Methodology

- A structured analytical methodology was adopted to address the selected problem statement. Initially, the provided raw datasets related to Data Science jobs, salaries, required skills, junior employee traits, and senior leadership personality factors were carefully reviewed to understand their structure, variables, and relevance to the study.



- The next step involved data preparation and preprocessing. Missing values were handled appropriately, duplicate records were removed, inconsistent entries were corrected, and important variables such as salary ranges, experience levels, and skill categories were standardized for accurate analysis. This process ensured that the datasets were reliable and ready for further exploration.
- After cleaning the data, Exploratory Data Analysis (EDA) was performed to identify major trends, patterns, and relationships. Different visual tools such as bar charts, pie charts, histograms, and comparative graphs were used to examine job demand, salary distribution, required skills, promotion factors, and leadership traits.
- Further analytical techniques were applied to compare technical skills with career growth outcomes, evaluate salary progression across roles and experience levels, and identify personality traits linked with success in senior positions. Correlation analysis and comparative methods were used wherever necessary.
- Finally, all insights obtained from the multiple datasets were integrated to build a unified and data-driven blueprint titled “From Entry-Level to Leadership”, highlighting the key factors that contribute to long-term success in Data Science careers.

Data Exploration

Dataset Summary

A comprehensive audit was performed across four distinct datasets to map the career progression from entry-level roles to leadership. This initial scan ensures the volume and variety of data are sufficient for building a reliable career blueprint.

Dataset Name	Total Records	Key Variables	Core Purpose
DataScience Jobs	1,602	Company, Role, Experience, Salary	Establishing market entry benchmarks
Analytics Jobs	15,841	Skills, Location, Salary Range	Mapping technical skill demand
JDS Skill Traits	139	Technical Skill Scores (1-5), Hike	Analyzing junior-level growth
SDS Personality Traits	139	OCEAN Traits, Success Metric	Identifying leadership success factors

Missing Values Overview

Understanding data gaps is essential for maintaining the integrity of the analysis. A systematic check for null values was conducted across all datasets:

Code Snippet

```
import pandas as pd
import os

os.makedirs('output', exist_ok=True)

try:
    df_jobs = pd.read_csv('DataScience Jobs.csv')
    print("Data successfully loaded!\n")
except FileNotFoundError:
    print("File nahi mili. Please check karo ki file same folder mein hai aur naam exact match kar raha hai.")

print("--- DATA INFO & MISSING VALUES ---")
print(df_jobs.info())
print("\nMissing Values Count:")
print(df_jobs.isnull().sum())

print("\n--- NUMERICAL DATA SUMMARY (MIN/MAX/MEAN) ---")
print(df_jobs.describe())

print("\n--- TOP JOB TITLES (Checking for typos) ---")
print(df_jobs['job_title'].value_counts().head(10))

with open('output/01_data_inspection.txt', 'w', encoding='utf-8') as f:
    f.write("--- DATA INFO & MISSING VALUES ---\n")
    f.write(df_jobs.info(buf=f))
    f.write("\nMissing Values Count:\n")
    f.write(df_jobs.isnull().sum().to_string())
    f.write("\n\n--- NUMERICAL DATA SUMMARY (MIN/MAX/MEAN) ---\n")
    f.write(df_jobs.describe().to_string())
    f.write("\n\n--- TOP JOB TITLES ---\n")
    f.write(df_jobs['job_title'].value_counts().head(10).to_string())

print("\nOutput saved to output/01_data_inspection.txt")
```

- **Analytics Jobs:** This dataset exhibits significant missing values in the job_description and job_type columns. However, critical features like key_skills and location are well-populated, allowing for robust extraction.
- **DataScience Jobs:** The dataset is entirely complete with no missing values, providing a solid foundation for salary and role analysis.
- **JDS & SDS Traits:** These behavioral and skill-based datasets are 100% complete with no missing entries, ensuring high reliability for correlation studies.

```

...  ✓ Data successfully loaded!

--- DATA INFO & MISSING VALUES ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1602 entries, 0 to 1601
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   reference_no           1602 non-null   int64
1   company_name           1602 non-null   object
2   job_title              1602 non-null   object
3   min_experience          1602 non-null   int64
4   avg_salary             1602 non-null   object
5   min_salary             1602 non-null   object
6   max_salary             1602 non-null   object
7   num_of_jobs            1602 non-null   int64
dtypes: int64(3), object(5)
memory usage: 100.2+ KB
None

Missing Values Count:
reference_no      0
company_name      0
job_title         0
min_experience     0
...
Data Architect           50
Name: count, dtype: int64

```

Initial Observations

- **Structural Variation:** Salary data is primarily stored as unstructured text (e.g., '12.8L' or '6to10'), which requires numerical conversion before any mathematical analysis.
- **Skill Clusters:** Job descriptions and skill columns often contain multi-value strings. This indicates a need for tokenization to identify which individual skills are most frequently requested by employers.
- **Formatting Anomalies:** The leadership dataset (SDS) was found to have inconsistent column headers with leading and trailing spaces, requiring a cleaning pass to ensure script stability.
- **Standardized Scoring:** Both the JDS and SDS datasets utilize normalized scoring systems (either 1-5 scales or 0-1 binary classifications), making them ideal for high-impact visual comparisons like heatmaps and boxplots.

Data Preparation and Cleaning Analysis

Actions Performed and Purpose:

- Salary values in text format such as "12.8L" or "₹15,00,000" were cleaned using regular expressions to extract numeric values only. This converted alphanumeric salary data into a usable numerical format for analysis.
- Three new columns were created: avg_salary_num, min_salary_num, max_salary_num, allowing direct calculations and comparisons.
- Rows containing negative experience values were identified and reviewed, as they may represent data entry errors.

```
+ Code + Markdown | ▶ Run All ⌂ Restart ⌂ Clear All Outputs | 📄 Jupyter Variables 📄 Outline ...

import numpy as np
import re

def extract_number(text):
    if pd.isna(text):
        return np.nan
    text = str(text).replace(',', '')
    numbers = re.findall(r'\d+\.?\d*', text)
    if numbers:
        return float(numbers[0])
    return np.nan

print("Cleaning Salaries...")

df_jobs['avg_salary_num'] = df_jobs['avg_salary'].apply(extract_number)
df_jobs['min_salary_num'] = df_jobs['min_salary'].apply(extract_number)
df_jobs['max_salary_num'] = df_jobs['max_salary'].apply(extract_number)

print("\n--- BEFORE VS AFTER CLEANING (First 5 Rows) ---")
print(df_jobs[['avg_salary', 'avg_salary_num']].head())

print("\n--- CLEANED NUMERICAL SUMMARY ---")
print(df_jobs[['min_experience', 'min_salary_num', 'max_salary_num', 'avg_salary_num']].describe())

negative_exp = df_jobs[df_jobs['min_experience'] < 0]
print(f"\nRows with negative experience: {len(negative_exp)}")

with open('output/02_data_cleaning.txt', 'w', encoding='utf-8') as f:
    f.write("--- BEFORE VS AFTER CLEANING (First 5 Rows) ---\n")
    f.write(df_jobs[['avg_salary', 'avg_salary_num']].head().to_string())
    f.write("\n--- CLEANED NUMERICAL SUMMARY ---\n")
    f.write(df_jobs[['min_experience', 'min_salary_num', 'max_salary_num', 'avg_salary_num']].describe().to_string())
    f.write(f"\nRows with negative experience: {len(negative_exp)}")

print("\n🟢 Output saved to output/02_data_cleaning.txt")
```

Key Insights from the Cleaned Data:

- The mean and range of minimum experience indicate that the market offers opportunities for both freshers and mid-level professionals.
- The mean and standard deviation of average salary help understand salary distribution and whether the market has equal pay levels or a wide compensation gap.

```
• Cleaning Salaries...

--- BEFORE VS AFTER CLEANING (First 5 Rows) ---
  avg_salary  avg_salary_num
0      7.8L             7.8
1     12.8L            12.8
2     13.4L            13.4
3      9.8L             9.8
4      8.6L             8.6
```

```

--- CLEANED NUMERICAL SUMMARY ---
      min_experience  min_salary_num  max_salary_num  avg_salary_num
count      1602.000000      1602.000000      1602.000000      1602.000000
mean         2.799001         8.634145        19.143258        13.234894
std          2.353686         5.796976        11.150180         7.840744
min           0.000000         0.200000         2.000000         1.400000
25%           1.000000         4.400000        12.000000         7.600000
50%           2.000000         7.000000        18.000000        11.900000
75%           4.000000        11.500000        24.000000        17.175000
max           21.000000        55.000000       102.000000        82.000000

```

Rows with negative experience: 0

✅ Output saved to output/02_data_cleaning.txt

Practical Applications:

- Salary Benchmarking – Compare offered salaries with market averages.
- Machine Learning Readiness – Cleaned data can be directly used for regression or prediction models.
- Error Detection – Negative experience rows should be removed to improve model accuracy.
- HR Decision Support – Organizations can use this data to set realistic salary bands based on market trends.

Data Science Jobs Visualization Analysis

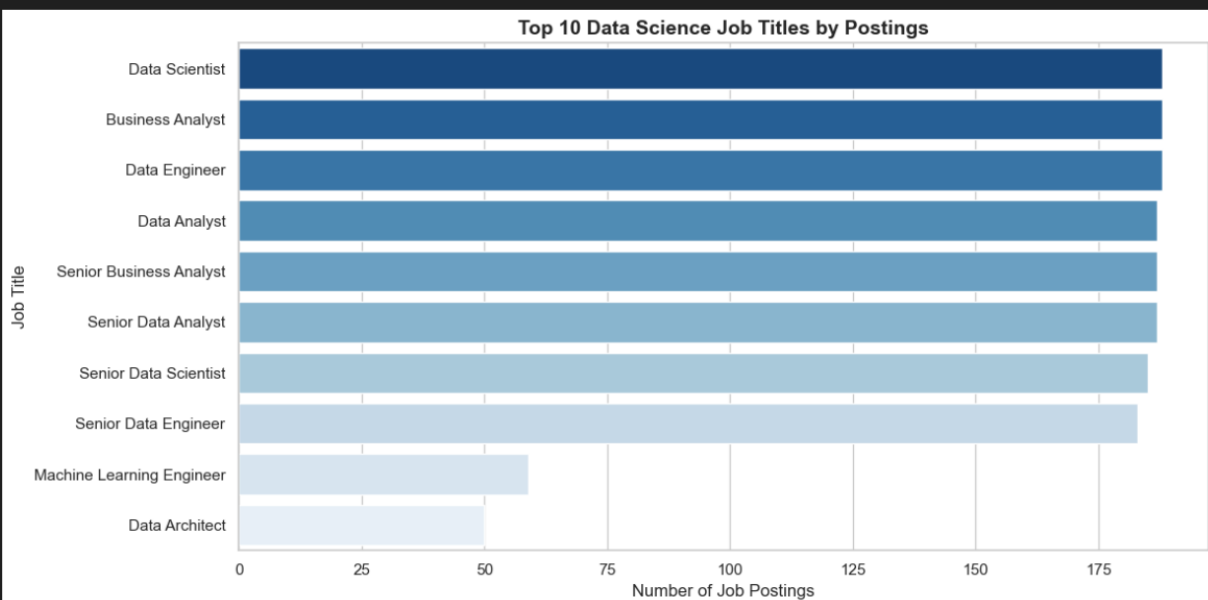
Graph 1 — Top 10 Job Titles

```

Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'y' variable to 'hue' and set 'legend=False' for the same effect.

sns.barplot(y=top_titles.index, x=top_titles.values, palette='Blues_r')

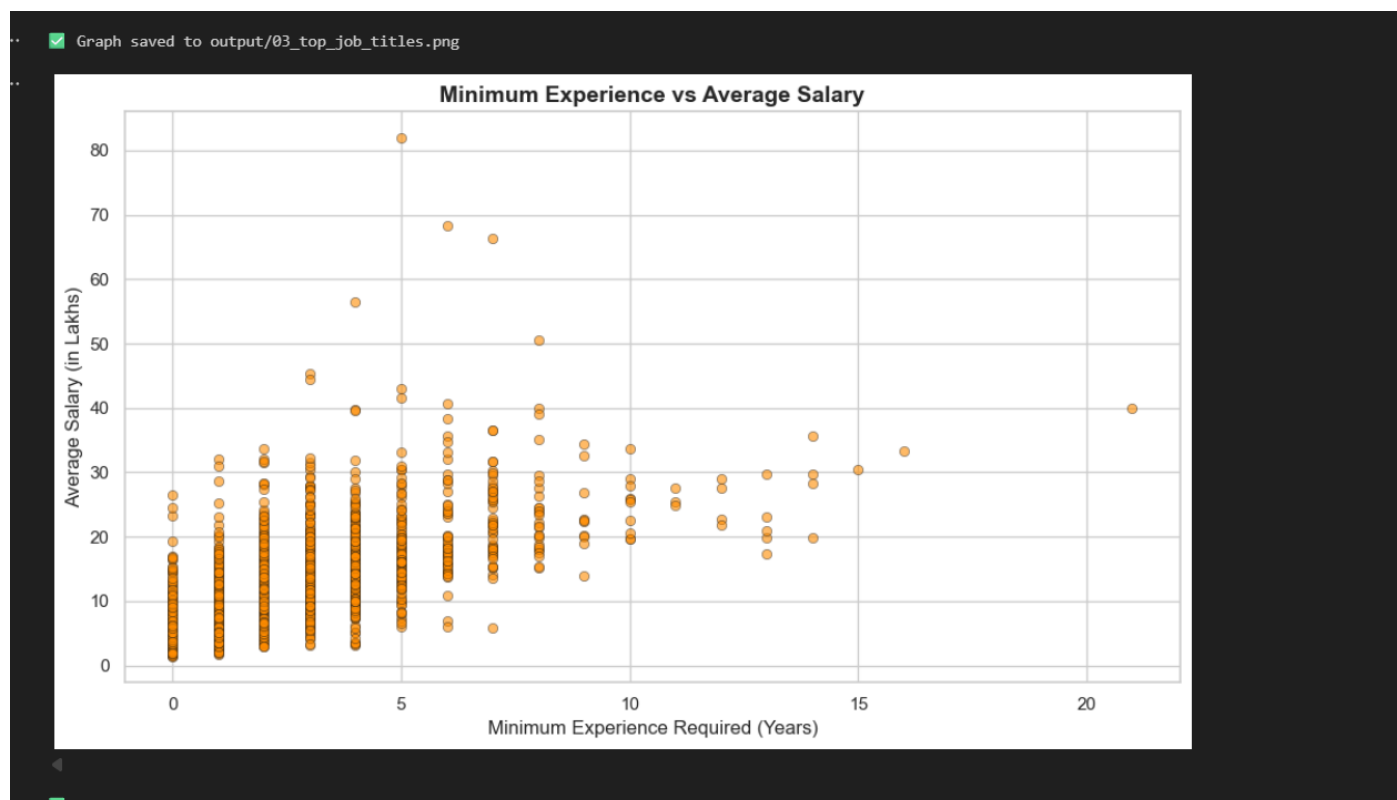
```



✅ Graph saved to output/03_top_job_titles.png

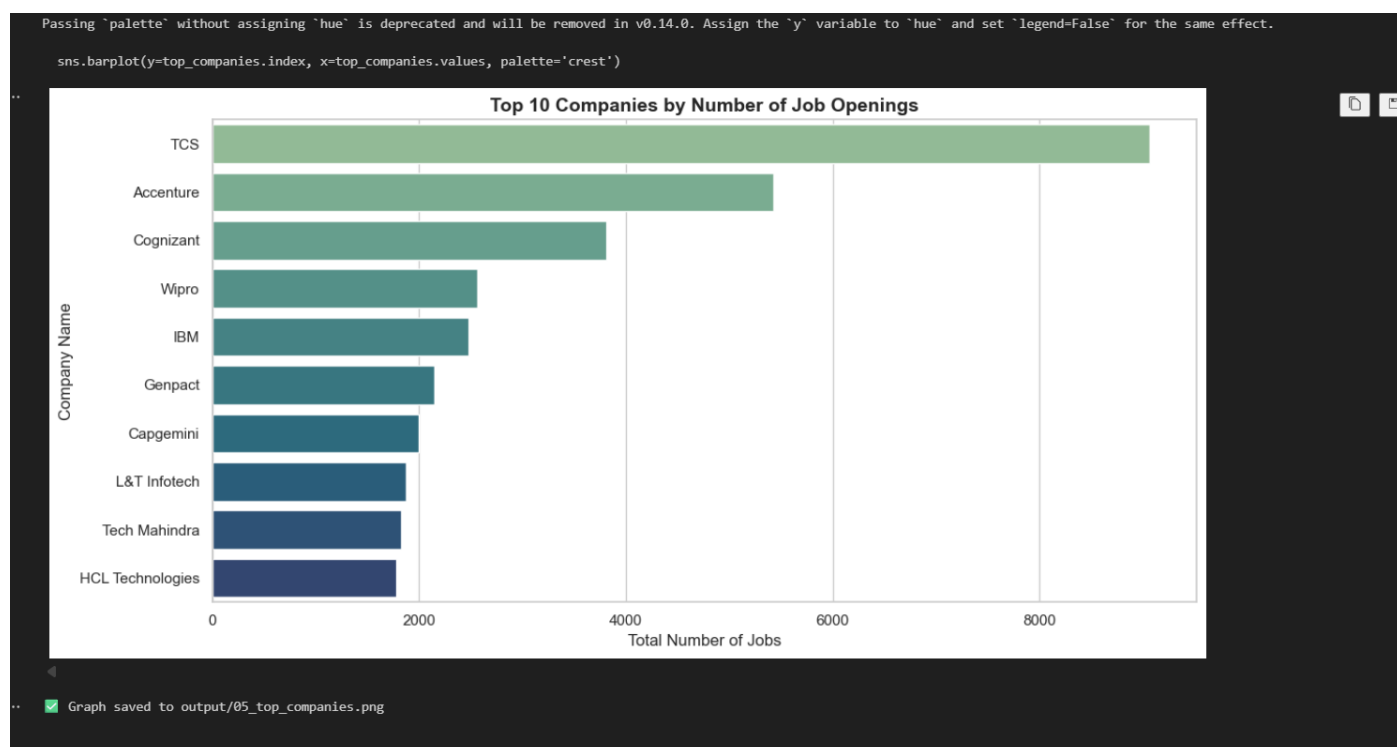
- The most frequently listed job roles indicate the highest market demand in the Data Science industry.
- If roles such as Data Analyst and Data Scientist appear at the top, they can be considered strong and reliable career options for aspiring professionals.

Graph 2 — Experience vs Salary (Scatter Plot)



- A generally positive correlation is expected, where higher experience leads to higher salary levels.
- If some outliers are present (high salary with low experience), they may represent niche roles or candidates with specialized skills.
- Flat regions in the graph may indicate a salary ceiling, where experience alone is not enough and skill upgrades or role changes become necessary.

Graph 3 — Top Hiring Companies



- These companies are actively recruiting data professionals and can be considered priority targets for job seekers.
- The list of top hiring companies is also useful for colleges and placement teams to build hiring partnerships.

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

plt.figure(figsize=(12, 6))
top_titles = df_jobs['job_title'].value_counts().head(10)
sns.barplot(y=top_titles.index, x=top_titles.values, palette='Blues_r')
plt.title('Top 10 Data Science Job Titles by Postings', fontsize=14, fontweight='bold')
plt.xlabel('Number of Job Postings', fontsize=12)
plt.ylabel('Job Title', fontsize=12)
plt.tight_layout()
plt.savefig('output/03_top_job_titles.png', dpi=300, bbox_inches='tight')
plt.show()
print("✅ Graph saved to output/03_top_job_titles.png")

plt.figure(figsize=(10, 6))
sns.scatterplot(data=df_jobs, x='min_experience', y='avg_salary_num', alpha=0.6, color='darkorange', edgecolor='k')
plt.title('Minimum Experience vs Average Salary', fontsize=14, fontweight='bold')
plt.xlabel('Minimum Experience Required (Years)', fontsize=12)
plt.ylabel('Average Salary (in Lakhs)', fontsize=12)
plt.tight_layout()
plt.savefig('output/04_experience_vs_salary.png', dpi=300, bbox_inches='tight')
plt.show()
print("✅ Graph saved to output/04_experience_vs_salary.png")

plt.figure(figsize=(12, 6))
top_companies = df_jobs.groupby('company_name')['num_of_jobs'].sum().sort_values(ascending=False).head(10)
sns.barplot(y=top_companies.index, x=top_companies.values, palette='crest')
plt.title('Top 10 Companies by Number of Job Openings', fontsize=14, fontweight='bold')
plt.xlabel('Total Number of Jobs', fontsize=12)
plt.ylabel('Company Name', fontsize=12)
plt.tight_layout()
plt.savefig('output/05_top_companies.png', dpi=300, bbox_inches='tight')
plt.show()
print("✅ Graph saved to output/05_top_companies.png")
```

Practical Applications

- For Job Seekers – Apply directly to top hiring companies and align resumes with common job description requirements.
- For Career Counselors – Guide students toward the most in-demand and employable roles.
- For Salary Negotiation – Use the experience vs salary trend to estimate and justify expected compensation.

Analytics Jobs Initial Inspection

Key Observations

- The Analytics Jobs dataset contains columns such as key_skills and location, where values are comma-separated. This means a single row may contain multiple skills and multiple cities.
- The salary column is also stored in string format, such as "8 to 12", which requires cleaning before analysis.
- If missing values are present in the key_skills column, those rows may need to be removed or handled appropriately for accurate skill analysis.

```

try:
    df_analytics = pd.read_csv('Analytics Jobs.csv')
    print("✅ Data successfully loaded!\n")

    print("--- DATA INFO ---")
    print(df_analytics.info())

    print("\n--- MISSING VALUES ---")
    print(df_analytics.isnull().sum())

    print("\n--- FIRST 5 ROWS ---")
    print(df_analytics.head())

    with open('output/06_analytics_inspection.txt', 'w', encoding='utf-8') as f:
        f.write("--- DATA INFO ---\n")
        df_analytics.info(buf=f)
        f.write("\n--- MISSING VALUES ---\n")
        f.write(df_analytics.isnull().sum().to_string())
        f.write("\n\n--- FIRST 5 ROWS ---\n")
        f.write(df_analytics.head().to_string())

    print("\n✅ Output saved to output/06_analytics_inspection.txt")

except Exception as e:
    print(f"Error loading data: {e}")

```

[4] Python

```

✅ Data successfully loaded!

--- DATA INFO ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15841 entries, 0 to 15840
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   s_no                  15841 non-null  int64
1   experience             15841 non-null  object
2   job_description        12333 non-null  object
3   job_desig             15841 non-null  object
4   job_type               3830 non-null   object
5   key_skills             15840 non-null  object
6   location               15841 non-null  object
7   salary                 15841 non-null  object
dtypes: int64(1), object(7)
memory usage: 990.2+ KB
None

--- MISSING VALUES ---
s_no          0
experience     0
job_description  3508
job_desig      0
...
3              Pune    6to10
4  Mumbai, Bengaluru, Hyderabad, Pune, Delhi  10to15

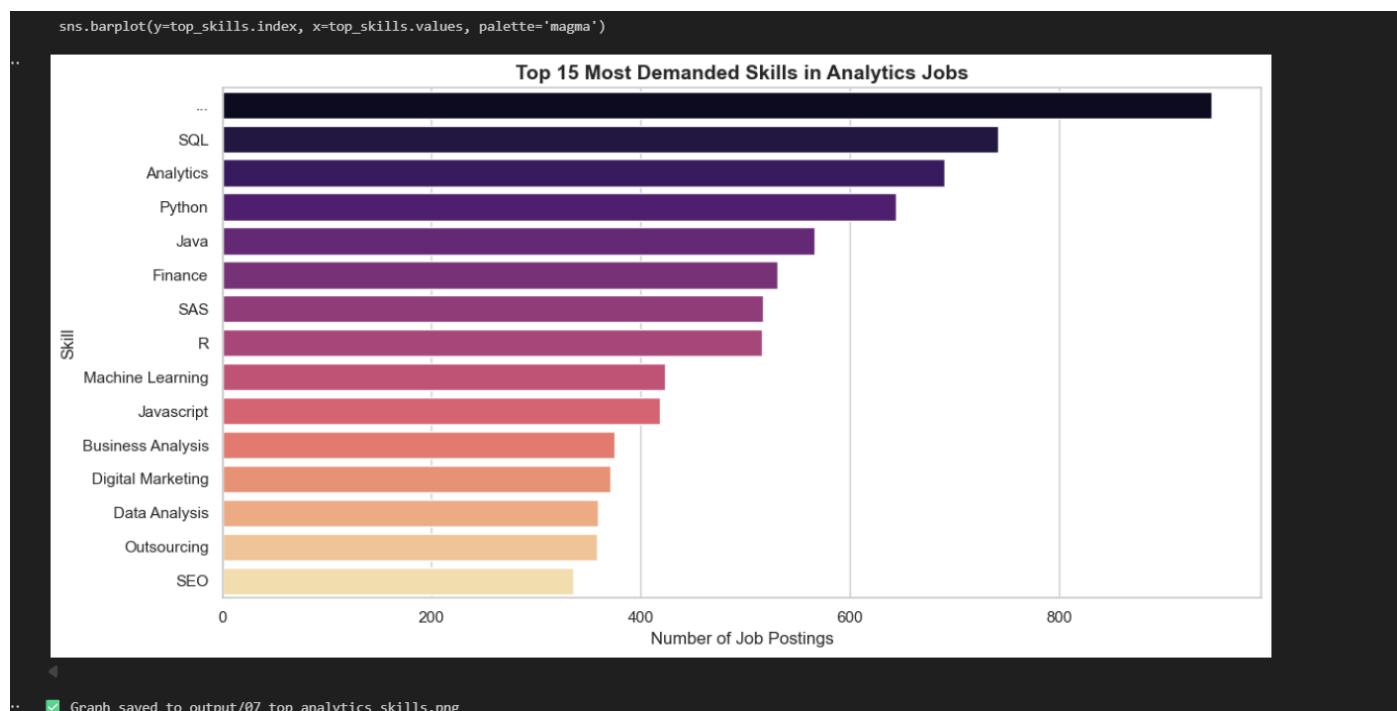
```

Practical Applications

- Skill Extraction – The dataset structure indicates that methods such as splitting or exploding values can be used to convert multiple skills into separate rows for better analysis.
- Location Analysis – City-wise demand for analytics jobs can be identified using the location column.
- Salary Standardization – Salary ranges can be cleaned and converted into numeric values for comparison and trend analysis.
- Data Understanding – Initial inspection helps in designing effective preprocessing steps before advanced analysis.

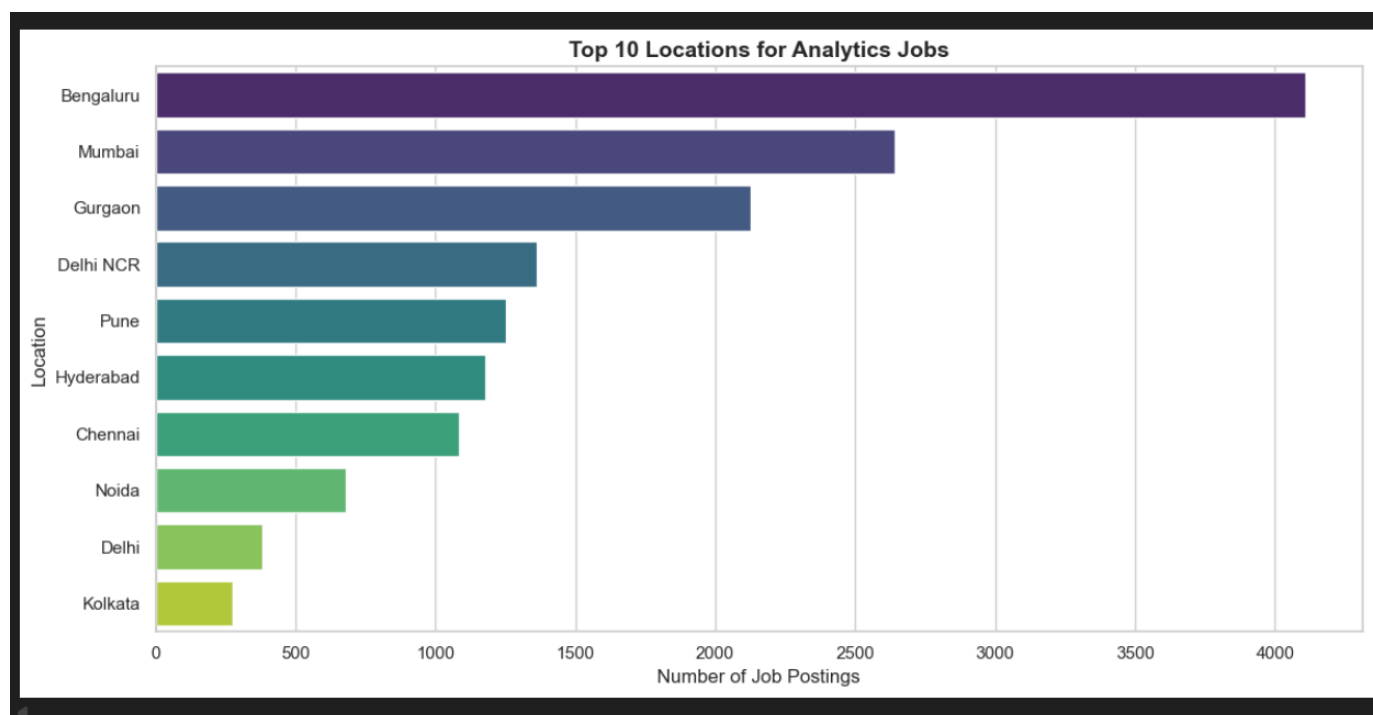
Analytics Jobs Visualization Analysis

Graph 1 — Top 15 Skills



- This graph highlights the **most in-demand technical skills** in the analytics job market.
- Skills such as **SQL, Excel, Python, and Tableau** often appear at the top, showing they are core requirements for many roles.
- This list can be used as a **learning roadmap**, where highly demanded skills should be prioritized first.

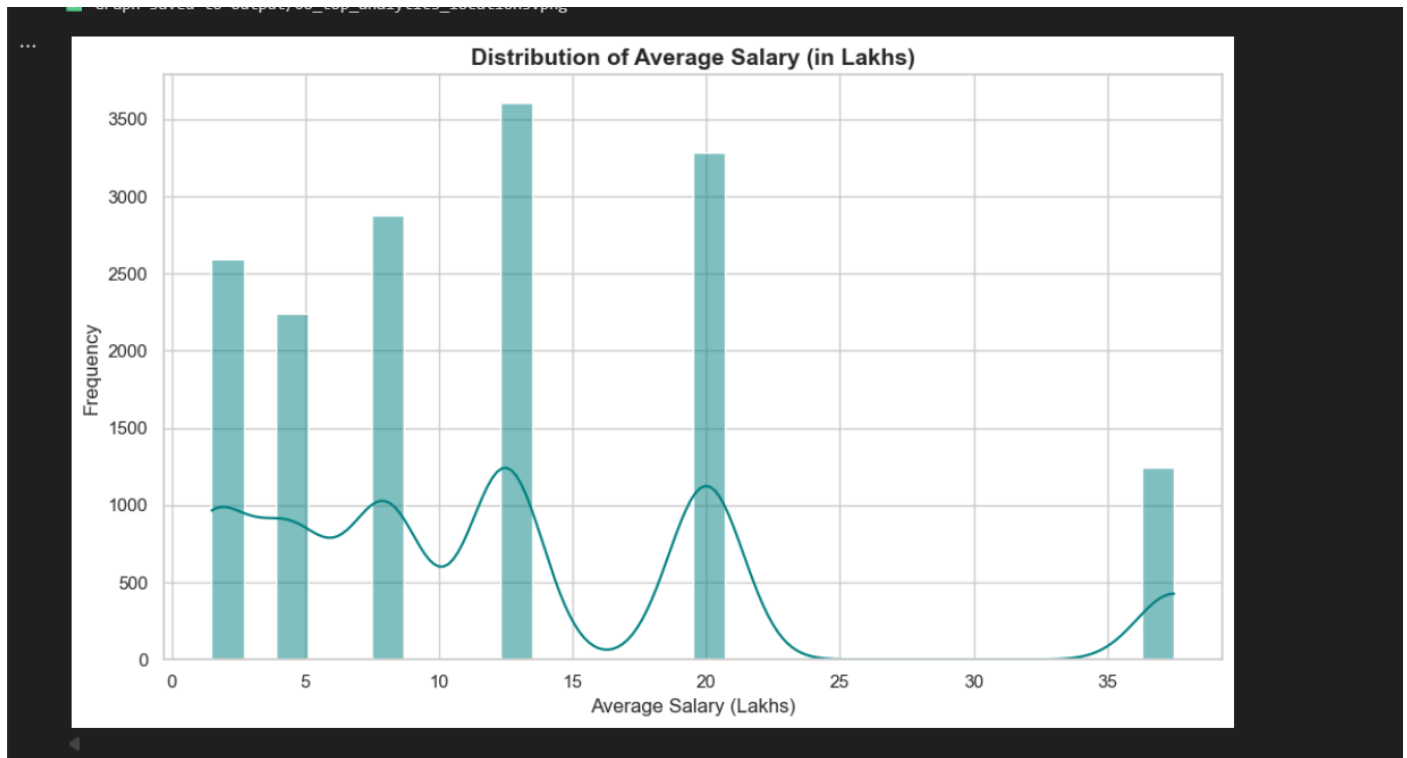
Graph 2 — Top Job Locations



- Cities such as **Bengaluru, Mumbai, and Hyderabad** generally show higher concentration of analytics job opportunities.

- Candidates willing to relocate can target **high-demand cities** for better job prospects.
- Remote opportunities may vary, so current job portals should also be reviewed for updated trends.

Graph 3 — Salary Distribution (Histogram + KDE)



- A **right-skewed distribution** suggests that most professionals fall in the low-to-mid salary range, while a smaller group earns significantly higher salaries.
- The KDE curve helps identify the **most common salary range**, useful for setting realistic expectations.
- If the graph shows two peaks, it may indicate dominance of both entry-level and senior-level roles, with fewer mid-level positions.

Practical Applications

- **Skill Gap Analysis** – Compare current skills with market demand to identify learning priorities.
- **Location Strategy** – Focus job search efforts on cities with stronger hiring demand.
- **Salary Negotiation** – Use salary distribution insights to evaluate whether an offer is competitive.

JDS Skill Traits Results Analysis

Dataset Overview

- This dataset evaluates candidates across five technical skill dimensions on a **1–5 scale: Big Data, Mathematics/Statistics, Coding, AI/ML, and Dashboarding/Storytelling**.
- The target variable **salary_hike_high_or_low** is binary, where **0 = low salary hike** and **1 = high salary hike**.

```

... --- INITIAL DATA INFO ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 139 entries, 0 to 138
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     139 non-null    int64
1   big_data_skills                       139 non-null    float64
2   maths-stats_skills                   139 non-null    float64
3   coding_skills                        139 non-null    float64
4   ai_and_ml_skills                     139 non-null    float64
5   dashboard_and_storytelling_skills    139 non-null    float64
6   salary_hike_high_or_low              139 non-null    int64
dtypes: float64(5), int64(2)
memory usage: 7.7 KB
None

--- SUMMARY STATISTICS ---

```

	id	big_data_skills	maths-stats_skills	coding_skills	ai_and_ml_skills	dashboard_and_storytelling_skills	salary_hike_high_or_low
count	139.000000	139.000000	139.000000	139.000000	139.000000	139.000000	139.000000
mean	2958.460432	3.849640	4.293525	4.268345	3.849640	3.849640	3.849640
std	604.538742	0.846594	0.843922	0.892971	0.846594	0.846594	0.846594
min	2007.000000	2.300000	2.200000	2.200000	2.300000	2.300000	2.300000
25%	2383.500000	3.100000	3.800000	3.350000	3.100000	3.100000	3.100000
50%	2963.000000	3.800000	4.600000	4.600000	3.800000	3.800000	3.800000

```

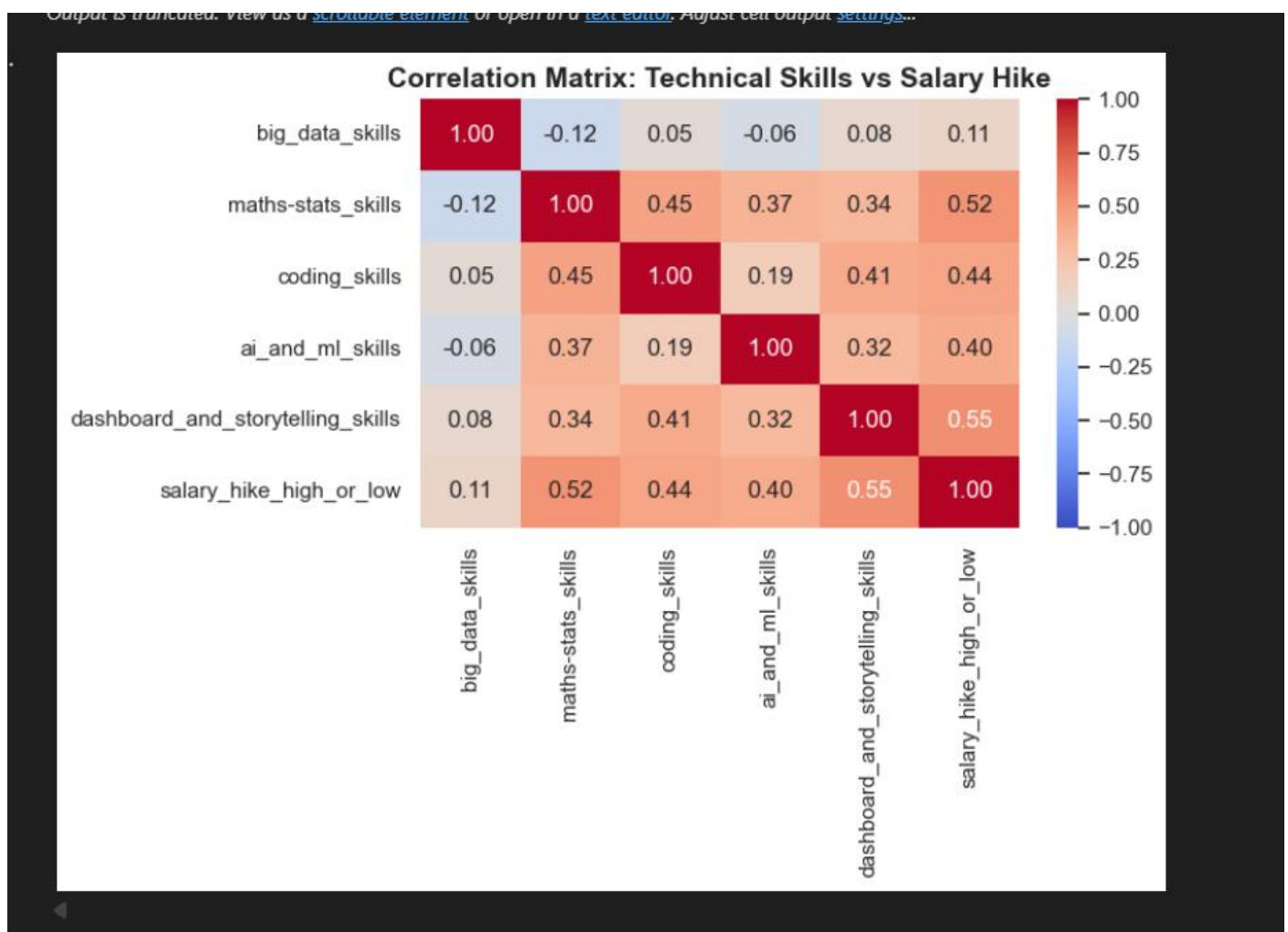
...
No outliers in ai_and_ml_skills.
No outliers in dashboard_and_storytelling_skills.

```

Outlier Treatment

- Some values were found outside the 1–5 scale and were clipped to the valid range.
- This step improves data quality, as Likert-scale datasets should remain within the defined limits.

Graph 1 — Correlation Heatmap



- Skills showing the **highest positive correlation** with **salary_hike_high_or_low** are likely the strongest drivers of salary growth.

- Inter-correlation among skills can also be observed. For example, if **AI/ML** and **Coding** are highly correlated, they may commonly develop together.

Graph 2 — Skills Comparison Bar Chart



Candidates receiving high salary hikes generally show stronger average scores across certain skills.

- The skill with the **largest difference** between low-hike and high-hike groups can be considered the most influential differentiator for career growth.

Practical Applications

- **Performance Reviews** – Managers can use skill scores to support compensation decisions.
- **Training Programs** – HR teams can design focused upskilling initiatives.
- **Self-Assessment** – Professionals can compare their skill levels with high-growth profiles.
- **Predictive Modeling** – This dataset is useful for building models that predict future salary hike potential.

SDS Personality Traits Inspection Analysis

Dataset Overview

- This dataset is based on the **OCEAN Model (Big Five Personality Theory)**, which includes **Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism**.
- The target variable **success_classification_high_low** indicates whether a candidate is professionally successful or not.
- Column names were cleaned by removing extra spaces and formatting inconsistencies, which is a common issue in spreadsheet datasets.

Summary Statistics Observations

- The average score of each personality trait is expected to be around the normalized midpoint. Any major deviation may indicate stronger dominance of a specific trait.
- Minimum and maximum values were reviewed to confirm that normalization and scaling were performed correctly.

```
Cleaned Columns: ['id', 'neuroticism', 'extraversion', 'openness_to_experience', 'agreeableness', 'conscientiousness', 'success_classification_high_low']

--- INITIAL DATA INFO ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 161 entries, 0 to 160
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     161 non-null    int64
1   neuroticism                           161 non-null    int64
2   extraversion                           161 non-null    int64
3   openness_to_experience                  161 non-null    int64
4   agreeableness                           161 non-null    int64
5   conscientiousness                       161 non-null    int64
6   success_classification_high_low         161 non-null    int64
dtypes: int64(7)
memory usage: 8.9 KB
None

--- SUMMARY STATISTICS ---
      id  neuroticism  extraversion  openness_to_experience \
count  161.000000    161.000000    161.000000             161.000000
mean   8484.596273    36.192547    43.204969             41.329193
std     280.357503    11.268050    12.134722             11.322420
min     8001.000000    17.000000    17.000000             18.000000
...
success_classification_high_low  0
dtype: int64
```

Practical Applications

- **Hiring Decisions** – Psychometric traits can be linked with future professional success for better hiring choices.
- **Team Composition** – Candidates with high conscientiousness and low neuroticism may perform better in deadline-driven environments.
- **Leadership Development** – Personality insights can help identify employees with strong long-term leadership potential.
- **Combined Analysis** – This dataset can be integrated with technical skills data to understand how personality and skills together drive success.

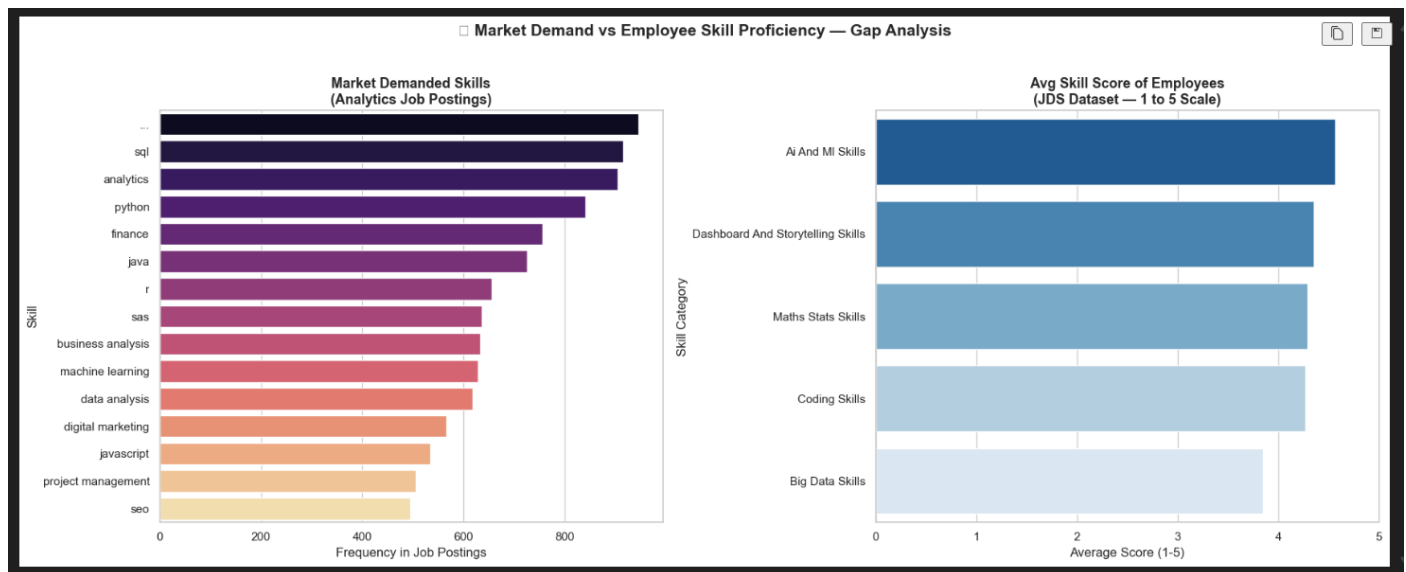
Skill Gap Analysis: Market Demand vs Employee Capability

Cross-Dataset Insights

- The left graph (Analytics Jobs dataset) represents **market demand**, showing which skills are most frequently requested in job postings.
- The right graph (JDS dataset) represents **employee capability**, highlighting the average proficiency levels of professionals across different skill categories.
- Comparing both graphs helps identify **critical skill gaps** where employer demand is high but employee proficiency is relatively low.
- For example, if **AI/ML** demand is high while employee AI/ML skill scores are lower, it indicates a major upskilling opportunity in the market.

Key Findings

- Skills with high demand and low capability should be considered priority areas for training and hiring.
- Skills with both high demand and high capability indicate mature talent availability.
- Low-demand but high-capability skills may suggest oversupply or underutilized talent.



Practical Applications

- **Training ROI** – Organizations can invest in the most impactful skill development areas.
- **Hiring Strategy** – If internal capability gaps are large, external hiring may be more effective than internal training.
- **Education Planning** – Colleges and bootcamps can align curriculum with actual market needs.
- **Career Growth** – Professionals can focus on high-demand skills to improve employability and salary potential.

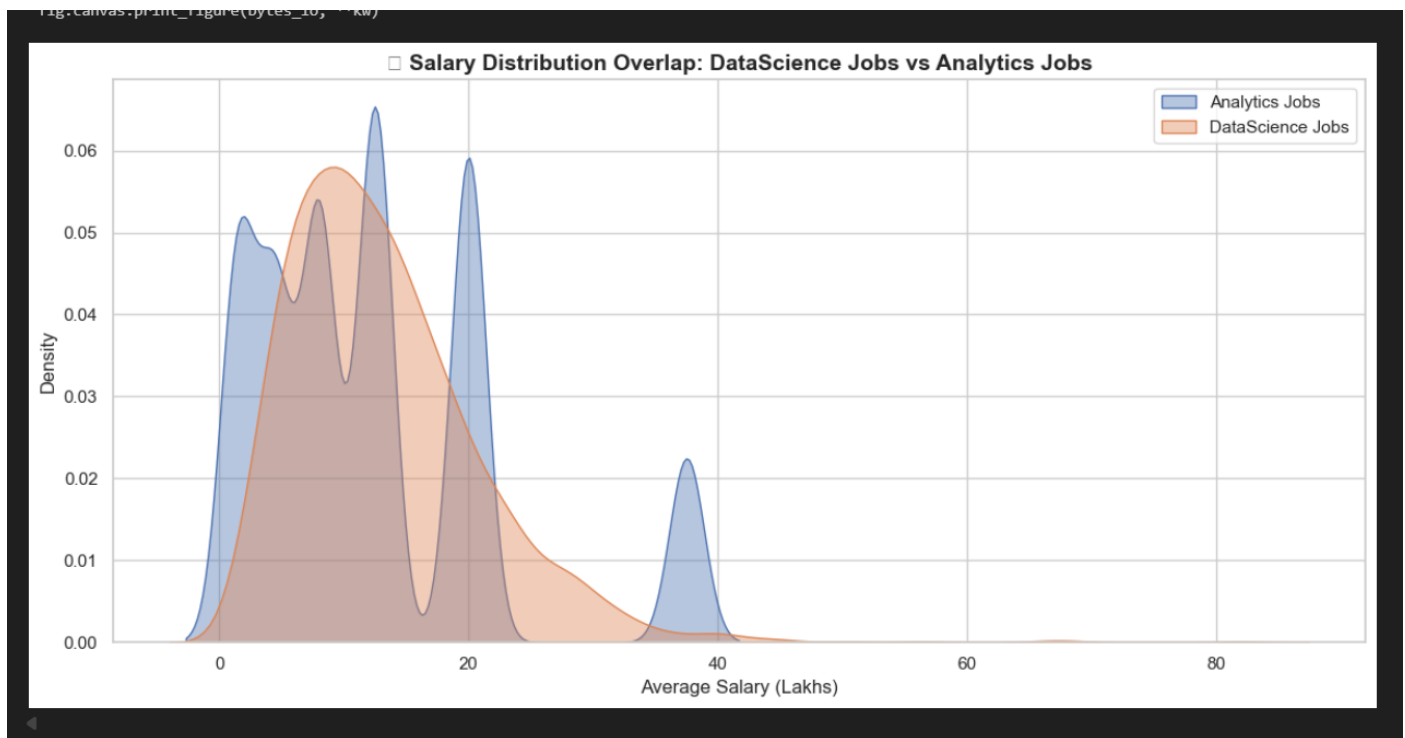
Salary Convergence Between Two Job Markets

KDE Overlap Analysis

- The KDE overlap graph compares the salary distributions of **Data Science** and **Analytics** job markets.
- If both curves significantly overlap, it suggests that the salary structures of the two fields are relatively similar.
- If the Data Science curve is shifted toward the right side, it indicates that Data Science roles generally offer higher compensation.
- Wider curves may represent greater salary variation within that job market.

Boxplot Comparison

- The boxplot allows direct comparison of median salaries between the two fields.
- The market with the **higher median salary** can be considered financially more rewarding on average.



- A wider **Interquartile Range (IQR)** indicates greater salary variance and stronger negotiation potential.
- Outliers may represent highly specialized or senior-level positions with premium compensation.

Key Findings

- Similar medians suggest close competition between the two career paths.
- Higher dispersion in one field may indicate more growth opportunities but also less consistency.
- Salary differences can help professionals choose between Analytics and Data Science based on career priorities.

Practical Applications

- **Career Decisions** – Candidates can select the field that better matches salary expectations and growth goals.
- **Recruiter Benchmarking** – Companies can compare their salary offers with competing markets.
- **Compensation Planning** – HR teams can maintain fair pay structures across Analytics and Data Science departments.
- **Upskilling Strategy** – Professionals may shift into higher-paying adjacent roles through targeted learning.

Ideal Candidate Analysis: Skills + Personality Combined

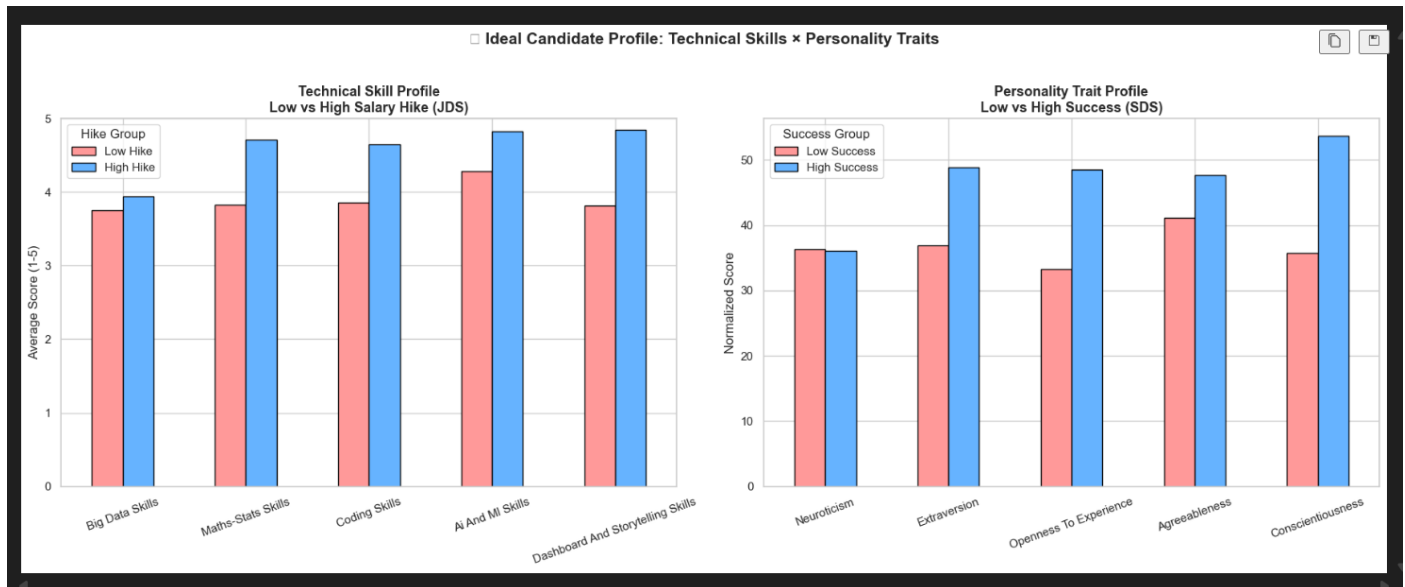
Key Insights from the Analysis

- The left graph compares the skill scores of candidates who received high salary hikes, highlighting the technical strengths linked to faster career growth.
- The right graph represents personality trait scores of candidates classified as professionally successful, showing behavioral traits associated with long-term success.

- A combined pattern suggests that candidates with strong Coding Skills and AI/ML Skills, along with high Conscientiousness and low Neuroticism, form an ideal success profile.

Cross-Dataset Validation

- Both datasets are independent sources, but if similar trends appear across them, the findings become more reliable and credible.
- If the JDS dataset shows coding as a major factor for salary hikes, and job market data also highlights Python/SQL as top skills, both datasets validate the same conclusion.
- This strengthens the argument that technical capability and personality together influence success in Data Science careers.



Key Findings

- Technical skills help accelerate short-term career growth and compensation.
- Personality traits support long-term consistency, leadership ability, and professional success.
- The strongest candidates are those who combine both technical excellence and positive behavioral traits.

Practical Applications

- Composite Candidate Scoring – Organizations can create a combined skill + personality score for hiring decisions.
- Learning & Development Strategy – Identify whether technical training or behavioral coaching is needed most.
- Recruitment Optimization – Improve candidate selection by evaluating both capability and mindset.
- Career Planning – Professionals can focus on building technical depth along with emotional and behavioral strengths

Results and Conclusions

Analytical Results and Key Insights

S.No	Analysis Category	Result / Finding	Clear Interpretation
1	Job Market Demand	Data Scientist and Data Analyst were the most demanded roles.	Best entry-level opportunities.
2	Most Required Skills	Python, SQL, Machine Learning, Excel, Tableau were top skills.	Students should focus on these first.
3	Average Salary	Data Science jobs showed higher average salary than Analytics jobs.	Better earning opportunities in DS.
4	Salary Growth Trend	Salaries increased with experience.	Growth improves with upskilling.
5	Salary Ceiling	Data Science roles had higher maximum salary packages.	Better long-term earning potential.
6	Junior Career Growth	Coding and AI/ML linked with salary hikes.	Technical depth drives promotions.
7	Leadership Success	High Conscientiousness and low Neuroticism linked with success.	Important for senior roles.
8	Skill Gap	AI/ML demand high but capability low.	Upskilling required.
9	Ideal Candidate	Strong Coding + AI/ML + Communication + Positive Traits.	Balanced candidates succeed faster.
10	Final Blueprint	Skills + Salary + Experience + Personality matter together.	Success is multi-dimensional.

Career Journey Summary

Career Stage	Salary Range	Key Requirement	Outcome
Entry-Level	Low to Moderate	Python, SQL, Excel	Higher hiring chances
Mid-Level	Moderate to High	Coding, AI/ML, Experience	Promotions & salary hikes
Senior-Level	High to Premium	Leadership Traits, Communication	Leadership success

The analysis of all provided datasets revealed clear patterns related to career growth in the Data Science field. It was found that **Data Scientist** and **Data Analyst** are the most demanded job roles in the market. Technical skills such as **Python, SQL, Machine Learning, and Data Visualization** were identified as the most valuable and frequently required skills.

Salary trends showed that professionals in Data Science roles generally earn higher compensation than those in Analytics roles, with salaries increasing as experience grows. Candidates with stronger **Coding** and **AI/ML** skills were more likely to receive faster promotions and better salary hikes.

Final Conclusion

Based on the analysis of multiple datasets related to jobs, salaries, skills, and personality traits, it can be concluded that success in Data Science careers is influenced by a combination of technical expertise, career experience, salary growth opportunities, and behavioral strengths.

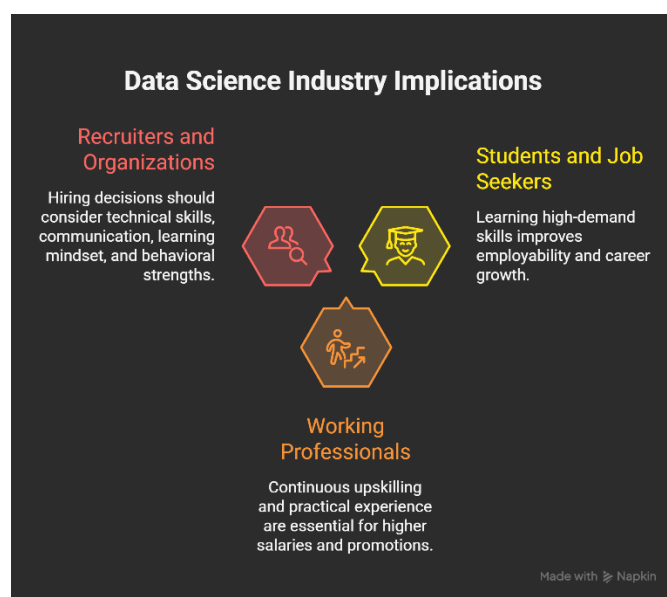
The study found that **Data Scientist** and **Data Analyst** roles are among the most demanded positions, while skills such as **Python**, **SQL**, **Machine Learning**, and **Data Visualization** remain highly valuable in the market. Salary growth generally increases with experience, but professionals with strong coding and AI/ML capabilities tend to achieve faster career progression.

In addition, personality traits such as **Conscientiousness**, **adaptability**, and **emotional stability** were strongly associated with long-term professional success and leadership potential. This shows that technical skills alone are not enough for sustained growth.

Therefore, a successful journey **from entry-level to leadership** requires continuous learning, upskilling, practical experience, and positive personality development. Individuals who balance both technical and personal competencies are more likely to achieve higher salaries, promotions, and leadership roles in the Data Science field.

Implications

- Students and job seekers should focus on learning high-demand skills such as **Python**, **SQL**, **Machine Learning**, and **Data Visualization**.
- Continuous upskilling and practical experience are important for better salary growth and promotions.



- Professionals should also develop soft skills such as communication, adaptability, and leadership qualities.
- Recruiters should evaluate candidates based on both technical skills and personality traits.

- Organizations can use these insights to design better training and employee development programs.
 - Companies should invest in closing skill gaps, especially in advanced areas like AI/ML.
 - The study provides a clear roadmap for building successful and sustainable Data Science careers.
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THANK YOU